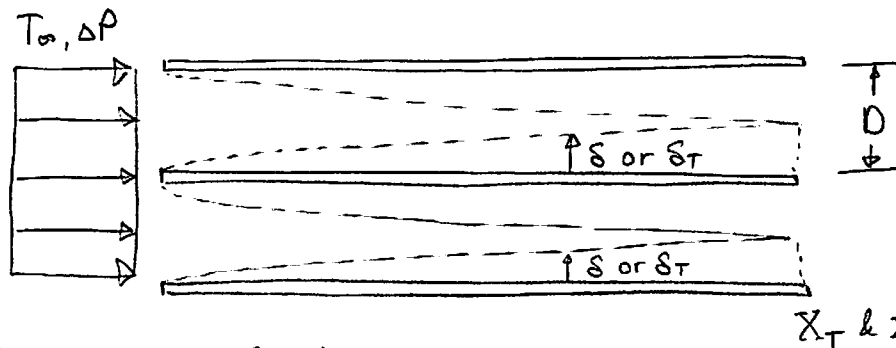


Large Spacing Limit: $D \rightarrow \infty$

For large spacing, each channel looks like the channel entrance region for the whole length:



Here, ΔP is fixed, so we need to solve for U_∞ that can achieve entrance effects for the whole plate.

A force balance on the whole control volume ($H \times L$) reveals:

$$\underbrace{\Delta P \cdot H}_{\text{Pressure Drop}} = \underbrace{n \cdot 2 \bar{\tau}_0 \cdot L}_{\text{Total shear force}} \Rightarrow \bar{\tau}_0 = \text{averaged shear stress over } L.$$

$$\bar{\tau}_0 = 1.328 Re_L^{-1/2} \cdot \frac{1}{2} \rho U_\infty^2$$

Back substituting:

$$U_\infty = \left(\frac{1}{1.328} \cdot \frac{\Delta P H}{n L^{1/2} \rho U_\infty^{1/2}} \right)^{2/3}$$

For the overall heat transfer rate from one board

$$\frac{\bar{h} L}{k} = \frac{\bar{q}''}{T_w - T_\infty} \cdot \frac{L}{k} = 0.664 Pr^{1/3} \left(\frac{U_\infty L}{\nu} \right)^{1/2} ; Pr > 0.5$$

$$q'_1 = \bar{q}'' \cdot L = k (T_w - T_\infty) 0.664 Pr^{1/3} \left(\frac{U_\infty L}{\nu} \right)^{1/2}$$

Assuming both sides are heating and maintained at T_w :

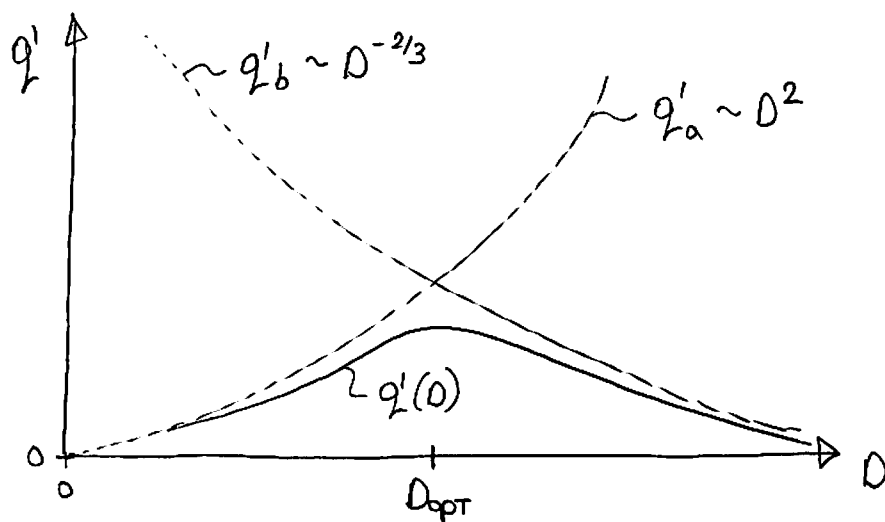
$$q'_{b'} = 2n q'_1 = 2nk (T_w - T_\infty) 0.664 Pr^{1/3} \left(\frac{U_\infty L}{\nu} \right)^{1/2}$$

But we know that $n = \frac{H}{D}$, and we solved for U_∞

$$q'_b = 1.208 k (T_w - T_\infty) H \frac{\rho_r^{1/3} L^{1/3} \Delta P^{1/3}}{\mu^{1/3} U^{2/3} D^{2/3}}$$

We see that $q'_b \sim D^{-2/3}$ in the large spacing limit.

This problem is a classical example of solution via intersection of asymptotes. We can do this since our two solutions are limiting cases only, and in between, mixed behaviour will occur:



To obtain our solution, we can equate the two solutions via scaling

$$q'_a \sim q'_b \Rightarrow \rho H \frac{D^2}{12\mu} \cdot \frac{\Delta P}{L} c_p (T_w - T_\infty) \sim 1.08 k (T_w - T_\infty) H \frac{(\rho_r L \Delta P)^{1/3}}{\mu^{1/3} U^{2/3} D^{2/3}}$$

$$D_{opt} \approx 2.73 L Be_L^{-1/4} \quad \text{for } 0.7 < Pr < 10^3$$

$$Be_L = \frac{\Delta P L^2}{\mu \alpha} = \text{Bejan \# (Dimensionless } \Delta P)$$

Note: $D_{opt, exp} = 3.05 L Be_L^{-1/4}$.

Results show for this solution that the board length (L) is of the same order of magnitude as the thermal entrance length (X_T).

Solving for our maximum heat transfer at D_{opt} , we obtain:

$$q'_{max} \leq 0.62 \left(\frac{p \Delta P}{\rho r} \right)^{1/2} \cdot H c_p (T_w - T_\infty) \Rightarrow T_w = \text{constant}$$

Two other cases that are useful to know are $q'' = \text{constant}$

$$D_{opt} = 3.2 L Be_L^{-1/4} \Rightarrow q'' = \text{constant}$$

$$q'_{max} \leq 0.4 \left(\frac{p \Delta P}{\rho r} \right)^{1/2} \cdot H \cdot c_p (T_{w,L} - T_\infty) ; T_{w,L} = T_w(L) \Rightarrow \text{max temp.}$$

For one side of each board $T_w = \text{constant}$, and the other adiabatic

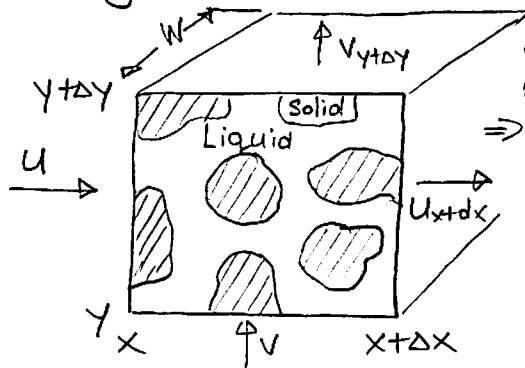
$$D_{opt} = 2.10 L Be_L^{-1/4}$$

$$q'_{max} \leq 0.37 \left(\frac{p \Delta P}{\rho r} \right)^{1/2} H c_p (T_w - T_\infty) \Rightarrow T_w = \text{constant on 1 side} \\ q'' = 0 \text{ on other.}$$

Convection in Porous Media

Relatively old problem due to need to manage the water table for irrigation systems.

Assuming we have a homogeneous porous medium:



⇒ We can estimate the flow to be 2D but locally it is always 3D in nature ⇒ just like turbulence

We assume $W \gg \Delta x$ and $W \gg \Delta y$ (2D flow). Only rates in the x & y directions are important:

$$\underbrace{\Delta x W}_{y \text{ cross-sect.}} \text{ and } \underbrace{\Delta y W}_{x \text{-cross sect.}} \gg \underbrace{\Delta x \Delta y}_{z \text{ cross section}}$$

$$\dot{m}_x = \rho \int_y^{y+\Delta y} \int_0^W u_p dz dy \Rightarrow u_p \equiv \text{uneven } x\text{-velocity distribution over void patches in } x\text{-plane.}$$

To make our lives simpler, we can determine the area averaged x -velocity:

$$u = \frac{1}{W \Delta y} \int_0^{y+\Delta y} \int_0^W u_p(y, z) dz dy \Rightarrow \dot{m}_x = \rho u (W \Delta y) \quad (1)$$

For the y -direction, we can do the same:

$$v = \frac{1}{W \Delta x} \int_0^{x+\Delta x} \int_0^W v_p(x, z) dz dx \Rightarrow \dot{m}_y = \rho v (W \Delta x) \quad (2)$$

Note, we've assumed that ρ is constant in the $\Delta x \Delta y$ domain, not necessarily over the entire x, y domain.

Applying mass conservation:

$$\frac{\partial M_{cv}}{\partial t} = \sum_{\text{inlet}} \dot{m} - \sum_{\text{outlet}} \dot{m} \quad (3) \Rightarrow M_{cv} \text{ is the instantaneous mass of the CV.}$$

We can define $M_{cv} = \rho W \Delta x \Delta y \phi \quad (4)$

$$\phi \equiv \text{porosity or void fraction} = \frac{\text{void volume}}{\text{total volume}}$$

Combining (3) & (4), we obtain:

$$\frac{\partial}{\partial t} (\rho \phi W \Delta x \Delta y) + \frac{\partial \dot{m}_x}{\partial x} \Delta x + \frac{\partial \dot{m}_y}{\partial y} \Delta y + \text{H.O.T.}(\Delta x^n, \Delta y^n) = 0$$

Back substituting (1) and (2)

$$\frac{\partial}{\partial t} (\rho \phi W \cancel{\Delta x \Delta y}) + \frac{\partial}{\partial x} (\rho u W \cancel{\Delta y}) \cancel{\Delta x} + \frac{\partial}{\partial y} (\rho v W \cancel{\Delta x}) \cancel{\Delta y} + \text{H.O.T.} = 0$$

Divide through by $\Delta x \Delta y$ and let $\Delta x, \Delta y \rightarrow 0$; H.O.T. $\rightarrow 0$

$$\phi \frac{\partial \rho}{\partial t} + \frac{\partial (\rho u)}{\partial x} + \frac{\partial (\rho v)}{\partial y} = 0$$

In general: $\boxed{\phi \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{V}) = 0}$

$\mathbf{V} \equiv$ volume averaged velocity vector (u, v, w)

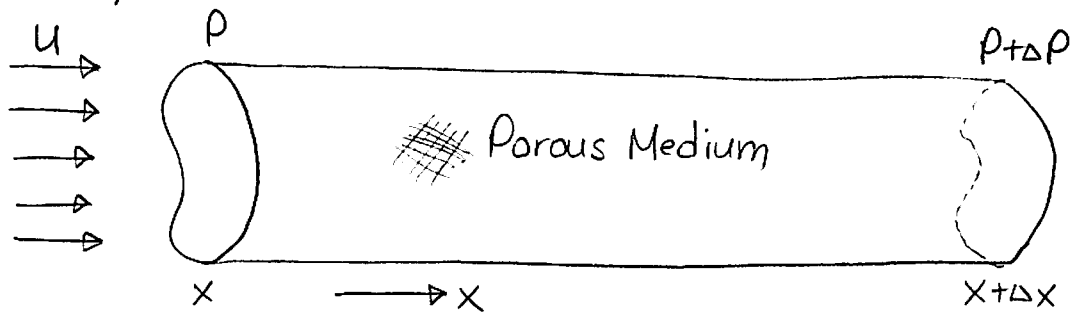
Note, if medium is a pure fluid, $\phi = 1$, and we return back to our original definition of mass conservation.

Darcy's Law

The main constitutive relation for flow through porous media is Darcy's law. Darcy, a french hydrologist empirically derived it by studying the flow in sand packed beds. It is the equivalent to Fourier's law in heat transfer:

$$u = \frac{K}{\mu} \left(-\frac{\partial p}{\partial x} \right) \quad (1), \quad K \equiv \text{permeability}$$

This law can be derived from Navier - Stokes equation, however it requires the permeability tensor and is skipped here.



Looking at the dimension of K :

$$[K] = \frac{[u][\mu]}{[-\partial p / \partial x]} = (\text{length})^2$$

Note the similarity of equation (1) & our Hagen - Poiseuille flow solution; For a pipe and channel

$$\bar{u} = \frac{r_0^2}{8\mu} \left(-\frac{\partial p}{\partial x} \right)$$

Pipe

$$\bar{u} = \frac{D^2}{12\mu} \left(-\frac{\partial p}{\partial x} \right)$$

Channel

$$K \sim r_0^2 \sim D^2$$

So: $K^{1/2} \equiv$ length scale of the pore diameter.

By assuming a small scale bundle of channels with H-P flow, we can derive (1) from it.

Defining our Reynolds number based on our pore scale:

$$\boxed{Re = \frac{\rho u K^{1/2}}{\mu}} \quad (2)$$

And our porous flow friction factor:

$$\boxed{f = \frac{\left(-\frac{\partial P}{\partial x}\right) K^{1/2}}{\rho u^2}} \quad (3) \Rightarrow \text{Before for H-P flow, we had } f = \frac{\left(-\frac{\Delta P}{L}\right) D^{1/2}}{\frac{1}{2} \rho u^2}$$

Back substituting our definition for u (1):

$$Re = \frac{\rho \frac{K}{\mu} \left(-\frac{\partial P}{\partial x}\right) K^{1/2}}{\mu} = \frac{\rho K^{3/2} \left(-\frac{\partial P}{\partial x}\right)}{\mu^2} \quad (4)$$

$$f = \frac{\left(-\frac{\partial P}{\partial x}\right) K^{1/2}}{\rho \left(\frac{K^{3/2} \left(-\frac{\partial P}{\partial x}\right)}{\mu^2}\right)} = \frac{\mu^2}{\rho K^{3/2} \left(-\frac{\partial P}{\partial x}\right)} \quad (5)$$

We see that: $\boxed{f = \frac{1}{Re}} \Rightarrow \text{Second form of Darcy's Law.}$

$\hookrightarrow$ Valid for laminar flow $\Rightarrow Re \leq 10$

Note, for $Re > 10$, inertia becomes important and we can use the Forchheimer modification:

$$\boxed{-\frac{\partial P}{\partial x} = \frac{\mu}{K} u + b \rho |u| u}$$

$b \equiv$ empirical constant based on geometry of pores.

From this stems:

$$\boxed{f = \frac{1}{Re} + 0.55} \Rightarrow Re > 10$$

If we have gravity present (body force = ρg_x)

$$u = \frac{K}{\mu} \left(-\frac{\partial p}{\partial x} + \rho g_x \right) \Rightarrow \text{Note, } u=0 \text{ when } \frac{\partial p}{\partial x} = \rho g_x$$

Pressure matches hydrostatics.

or in 3D:

$$\boxed{\mathbf{V} = \frac{K}{\mu} (-\nabla p + \rho \mathbf{g})}; \quad \mathbf{V} = (u, v, w)$$

$$\mathbf{g} = (g_x, g_y, g_z)$$

In many typical problems involving seepage flow of water in the ground, ρ and $\mu = \text{constant}$, $\mathbf{g} = (0, -g, 0)$

$$\mathbf{V} = -\frac{K}{\mu} \nabla \mathcal{E} \Rightarrow \mathcal{E} = p + \rho g y$$

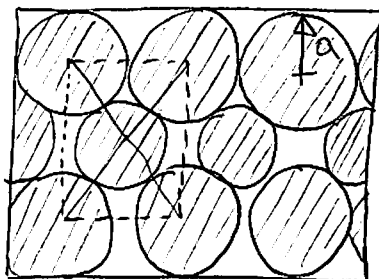
Our mass conservation for $\rho = \text{const.}$ becomes: $\nabla \cdot \mathbf{V} = 0$
Combining, we obtain:

$$\boxed{\nabla^2 \mathcal{E} = 0} \Leftrightarrow \boxed{\nabla^2 T = 0} \Leftrightarrow \boxed{\nabla^2 E = 0}$$

Can solve the seepage problem using steady state heat conduction.

Note, one important thing to discuss is the difference between permeability and porosity.

Spheres Diameter a



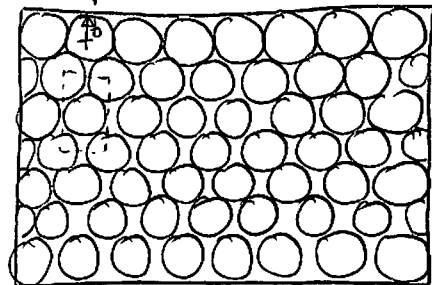
$$\phi_a \equiv \text{porosity} = \frac{2(\pi a^2)}{(2a)(3.464a)}$$

$$\phi_a = 0.91$$

$$K_a \approx a$$

or

Spheres Diameter b



$$\phi_b \equiv \text{porosity} = \frac{2(\pi b^2)}{(2b)(3.464b)}$$

$$\phi_b = 0.91 = \phi_a$$

$$K_b \approx b$$